

intrinsic plausibility

HOW GOOD IS A HYPOTHESIS? PLAUSIBILITY — Now to define intrinsic plausibility, also known as a **regularizer**. We find a hypothesis more plausible when its “total amount of dependence” on the features is small. So we’ll focus for now on capturing this intuition: *a hypothesis that depends a lot on many features is less plausible*.[◦] We may conveniently quantify this as proportional to a sum of squared weights (jargon: **L2**):[◦] implausibility of $h = (a, b, \dots) = \lambda(a^2 + b^2 + \dots)$. In code:

```
LAMBDA = 1.
def implausibility(a,b):
    return LAMBDA * np.sum(np.square([a,b]))
```

Intuitively, the constant $\lambda = \text{LAMBDA}$ tells us how much we care about plausibility relative to goodness-of-fit-to-data.

Here’s what the formula means. Each of three friends has a theory[◦] about which birds sing. Which theory do we prefer? Well, **AJ** seems too confident. Wingspan may matter but probably not so decisively. **Pat** avoids black-and-white claims, but Pat’s predictions depend substantively on many features: flipping any one quality flips their prediction. This seems implausible. By contrast, **Sandy**’s hypothesis doesn’t depend too strongly on too many features. To me, a bird non-expert, Sandy’s seems most plausible.

Now we can define the overall undesirability of a hypothesis:[◦]

```
def objective_function(examples,a,b):
    data_term = np.sum([svm_loss(x,y,a,b) for x,y in examples])
    regularizer = implausibility(a, b)
    return data_term + regularizer
```

MARGINS — To build intuition let’s suppose λ is a tiny positive number. Then minimizing the objective function is the same as minimizing the data term, the total SVM loss: our notion of implausibility breaks ties.

Now, how does it break ties? Momentarily ignore the Figure’s rightmost **orange point** and consider the black hypothesis; its predictions depend only on an input’s first (vertical) coordinate, so it comes from weights of the form $(a, b) = (a, 0)$. The $(a, 0)$ pairs differ in SVM loss. If $a \approx 0$, each point has leeway close to 0 and thus SVM loss close to 1; conversely, if a is huge, each point has leeway very positive and thus SVM loss equal to the imposed floor: 0. So SVM loss is 0 as long as a is so big that each leeway to exceed 1.

Imagine sliding a point through the plane. Its leeway is 0 at the black line and changes by a for every unit we slide vertically. So the farther the point is from the black line, the less a must be before leeway exceeds 1 — and the happier is the regularizer, which wants a small. So *minimizing SVM loss with an L2 regularizer favors decision boundaries far from even the closest correctly classified points!* The black line’s margins exceed the gray’s, so we favor black.

For large λ , then this margin-maximization tendency can be so strong that it overrides the data term. Thus, even when we bring back the rightmost **orange point** we ignored, we might prefer the black hypothesis to the gray one.

By the end of this section, you’ll be able to

- explain how regularization, in its incarnation as margin-maximization, counters data terms to improve generalization
- write a regularized ML program (namely, an SVM), to classify high-dimensional data

← There are many other aspects we might design a regularizer to capture, e.g. a domain’s symmetry. The regularizer is in practice a key point where we inject domain knowledge.

← **Food For Thought:** When (a, b) represent weights for brightness-width digits features, how do hypotheses with small $a^2 + b^2$ visually differ from ones with small $6.86a^2 + b^2$ (a perfectly fine variant of our ‘implausibility’)?

← **AJ** insists a bird with a wings shorter than 1ft can’t fly far, so it’s *sure* to sing; Conversely, birds with longer wings never sing. **Pat** checks if the bird grows red feathers, eats shrimp, lives near ice, wakes in the night, and has a bill. If and only if an even number of these 5 qualities are true, the bird probably sings. **Sandy** says shorter wings and nocturnality both make a bird somewhat more likely to sing.

← We’ll use SVM loss but feel free to plug in other losses to get different learning behaviors!

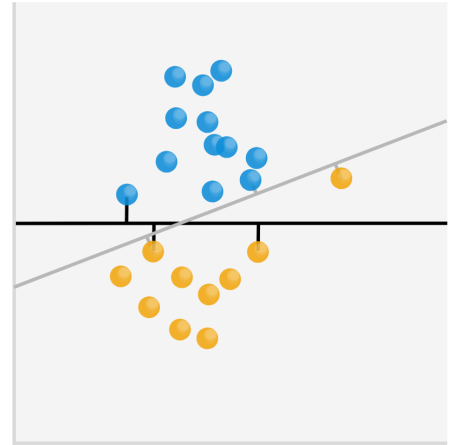


Figure 10: **Balancing goodness-of-fit against intrinsic plausibility leads to hypotheses with large margins.** A hypothesis’s **margin** is its distance to the closest correctly classified training point(s). Short stems depict these distances for two hypotheses (**black**, **gray**). If not for the rightmost **orange point**, we’d prefer **black** over **gray** since it has larger margins. With large λ (i.e., strong regularization), we might prefer black over gray even with that rightmost **orange point** included, since expanding the margin is worth the single misclassification.

Now for some really good intuition-building brain-food!

Food For Thought: Identify which point on the gray curve to the right corresponds to $\lambda = 0$. How about $\lambda = \infty$?

Food For Thought: We have two weight coefficients (corresponding to the horizontal and vertical axes of the Figure). Based on the **fit-to-data** term, which coefficient is the loss more sensitive to?

Food For Thought: Observe that the weight-vs- λ trajectory is curved: it doesn't interpolate linearly between its $\lambda = 0$ and $\lambda = \infty$ values. Which weight (horizontal or vertical) gets suppressed 'first' as we increase λ from 0?

Food For Thought: By thinking about points at which blue and orange contours are mutually tangent, sketch the weight-vs- λ trajectory described in the Figure. That is: check that the Figure is right!

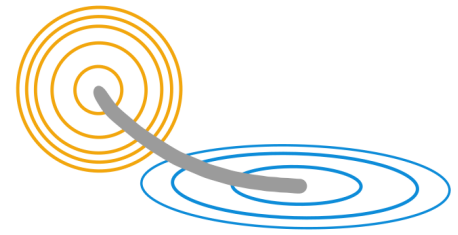


Figure 11: **Regularization suppresses different features by different amounts.** We show a contour plot of loss terms over 2D weight space: an **L2 regularizer** and a **fit-to-data** term. As we vary λ from 0 (**L2** doesn't matter) toward ∞ (**data** doesn't matter), the optimal weight changes. We show this weight-vs- λ trajectory in gray. **Warning:** For the perceptron and hinge notions of fit-to-data, the latter term won't look so smooth. Still, the moral about regularization applies. (And future models we'll discuss (logistic models, least-squares regression, etc) *are* smooth.)

OPTIMIZATION — Now that we've defined our objective function, we want to find a hypothesis $h = (a, b)$ that minimizes it. We've already discussed how to nudge the weight vector to reduce the badness-of-fit for a datapoint. How do we nudge it to reduce the implausibility? Well, we reduce the λ term simply by moving a, b closer to 0! That is, we combine an update of the form

$$w^{\text{new}} = w^{\text{old}} - \lambda w^{\text{old}}$$

with the data update.

(To get this to match our objective exactly, we should actually write $2\lambda/N$ instead of λ . The 2 comes from the second power in L2's definition; the $1/N$, more importantly, comes from the fact that we have N data terms but just 1 plausibility term. So if we work row-by-row (datapoint-by-datapoint), we ought to divvy up the plausibility term into N many terms, each of strength λ/N . At this point, we can just abstract this reasoning away by defining a new constant — say L — that secretly is $2\lambda/N$. Later, it'll be good to know where L comes from.)

We end up with

```
ETA = 0.01
ab = initialize()
for t in range(10000):
    xfeatures, y = fetch_datapoint_from(training_examples)
    ab = ab + ETA * ( - L * ab
                     + y * xfeats * (0 if max(0., y*ab.dot(xfeatures))==0 else 1) )
```

This is the **pegasos algorithm** we'll see in the project. Soon we'll formalize and generalize this algorithm using calculus.

Food For Thought: We've discussed the L2 regularizer. Also common is the L1 regularizer: implausibility of $h = (a, b, \dots) = \lambda(|a| + |b| + \dots)$. Hypotheses optimized with strong L1 regularization will tend to have zero dependence on many features. Explain to yourself and then to a friend what the previous sentence means, why it is true, and how we might exploit it in practice.



Figure 12: With $\lambda = 0.02$ the objective visibly prefers weights near 0. We develop an algorithm to take steps in this plane toward the minimum, 'rolling down' the hill so to speak.